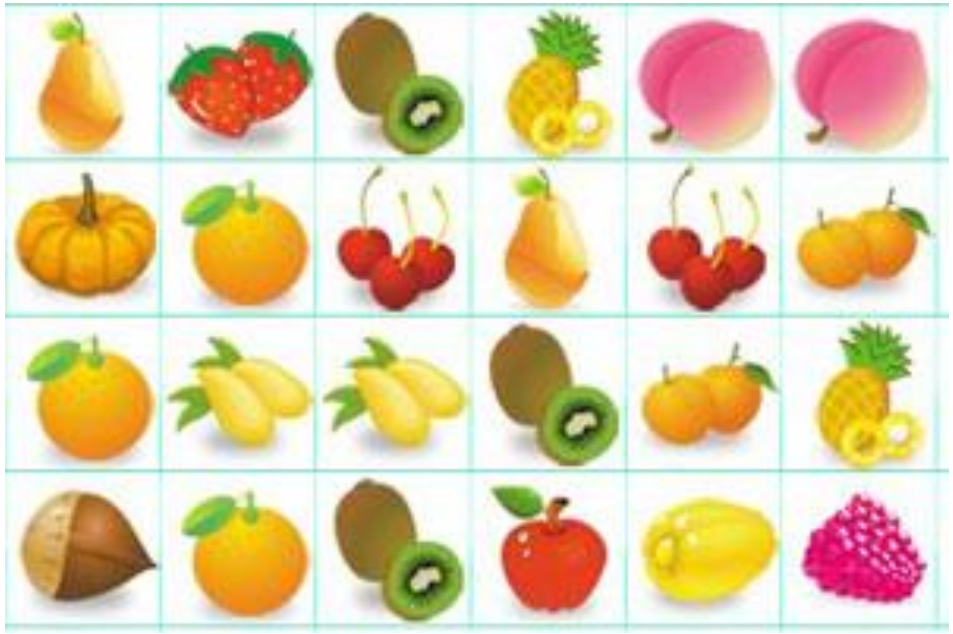
3. 卡诺图



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孙晓光

2026-10-12

1

卡诺图

卡诺图　Maurice Karnaugh (1924 -)

① **真值表**：优点：唯一。缺点：规模过大。

② **函数式**：不唯一。

各等价表达式之间有繁简之分，而无正错之分。



贝尔实验室

电信工程师

1953年

发明卡诺图

A	B	F
0	0	0
0	1	1
1	0	0
1	1	1

$\bar{A}B = 1$

$AB = 1$

定理：$F = \bar{A}B + AB = (\bar{A} + A)B = B$

相邻的**最小项**消去取值不同的因子后可以合并

A \ B	0	1
0	0	1
1	0	1

$\bar{A}B = 1$

$AB = 1$

2变量卡诺图

排列方案

不是唯一的

x \ y	0	1
0	00	01
1	10	11

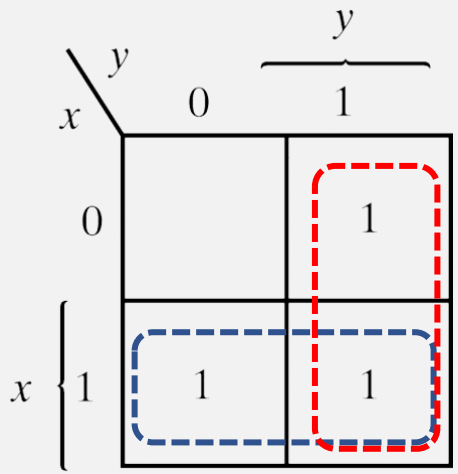
x \ y	0	1
0	$\bar{x}\bar{y}$	$\bar{x}y$
1	$x\bar{y}$	xy

x \ y	0	1
0	m_0	m_1
1	m_2	m_3

示例

x \ y	0	1
0	$\bar{x}\bar{y}$	$\bar{x}y$
1	$x\bar{y}$	xy

$$F = x\bar{y} + xy = x$$
$$= m_2 + m_3$$



$$F = x + y$$

3变量卡诺图-1

x \ yz	00	01	10	11
0	$\bar{x}\bar{y}\bar{z}$	$\bar{x}\bar{y}z$	$\bar{x}y\bar{z}$	$\bar{x}yz$
1	$x\bar{y}\bar{z}$	$x\bar{y}z$	$xy\bar{z}$	xyz

$$F = \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}yz$$

$$= \bar{y}\bar{z}(\bar{x} + x) + \bar{x}z(\bar{y} + y)$$

$$= \bar{y}\bar{z} + \bar{x}z$$

x \ yz	00	01	10	11
0	1	1	0	1
1	1	0	0	0

相邻：最小项相差一个变量。

x \ yz	00	01	11	10
0	1	1	1	0
1	1	0	0	0

格雷码

x	y	z	F	
0	0	0	1	$\bar{x}\bar{y}\bar{z} = 1$
0	0	1	1	$\bar{x}\bar{y}z = 1$
0	1	0	0	
0	1	1	1	$\bar{x}yz = 1$
1	0	0	1	$x\bar{y}\bar{z} = 1$
1	0	1	0	
1	1	0	0	
1	1	1	0	

K图竖过来如何呢？

3变量卡诺图-2

xy \ z	0	1
00	$\bar{x}\bar{y}\bar{z}$	$\bar{x}\bar{y}z$
01	$\bar{x}y\bar{z}$	$\bar{x}yz$
10	$x\bar{y}\bar{z}$	$x\bar{y}z$
11	$xy\bar{z}$	xyz

相邻：上下、左右也相邻

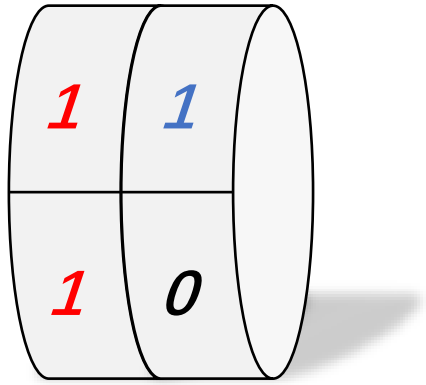
xy \ z	0	1
00	1	1
01	0	1
10	1	0
11	0	0

xy \ z	0	1
00	1	1
01	0	1
11	0	0
10	1	0

$$F = \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}yz$$

$$= \bar{y}\bar{z}(\bar{x} + x) + \bar{x}z(\bar{y} + y)$$

$$= \bar{y}\bar{z} + \bar{x}z$$



x	y	z	F	
0	0	0	1	$\bar{x}\bar{y}\bar{z} = 1$
0	0	1	1	$\bar{x}\bar{y}z = 1$
0	1	0	0	
0	1	1	1	$\bar{x}yz = 1$
1	0	0	1	$x\bar{y}\bar{z} = 1$
1	0	1	0	
1	1	0	0	
1	1	1	0	

3变量卡诺图-3

x \ yz	00	01	11	10
0	1	1	0	1
1	1	0	0	1

$$F = \bar{x}\bar{y} + \bar{y}\bar{z} + y\bar{z}$$

$$= \bar{x}\bar{y} + (\bar{y} + y)\bar{z}$$

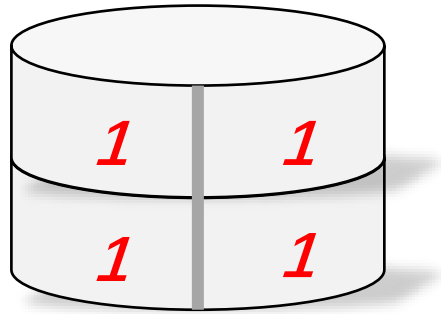
$$= \bar{x}\bar{y} + \bar{z}$$

单元格可以重复使用！$(A + A = A)$

x \ yz	00	01	11	10
0	1	1	0	1
1	1	0	0	1

合并相邻单元格，

需要找最大的合并！



【练习1】已知函数F的卡诺图，试化简

①

x \ yz	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$F = x\bar{y}\bar{z} + \bar{x}\bar{y}z + xyz + \bar{x}y\bar{z}$$

②

x \ yz	00	01	11	10
0	1	1	1	0
1	1	0	1	1

$$F = \bar{x}\bar{y} + yz + x\bar{z}$$

③

x \ yz	00	01	11	10
0	1	1	1	1
1	0	0	1	1

$$F = \bar{x} + y$$

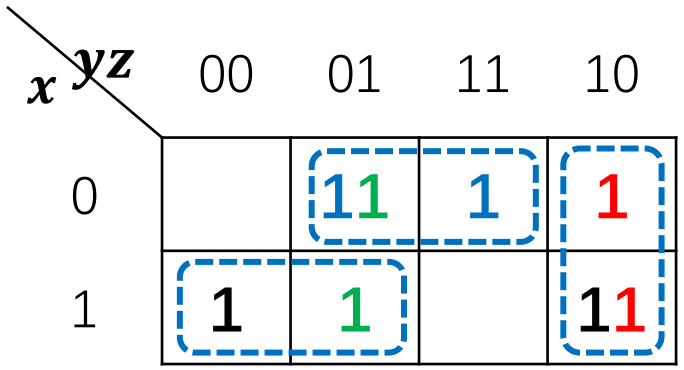
④

x \ yz	00	01	11	10
0	1	1	1	1
1	1	1	1	1

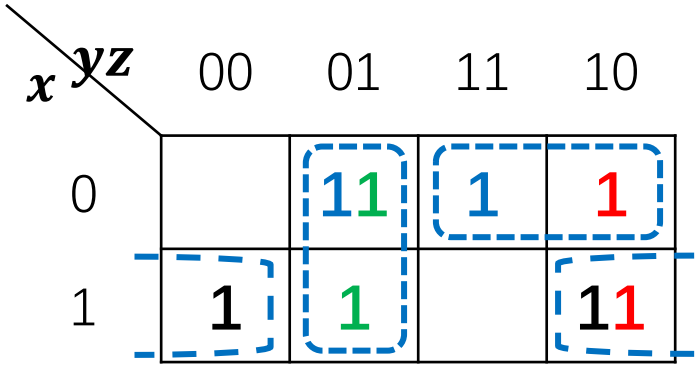
$$F = 1$$

【练习2】用卡诺图化简

① $F = x\bar{z} + \bar{x}z + y\bar{z} + \bar{y}z$

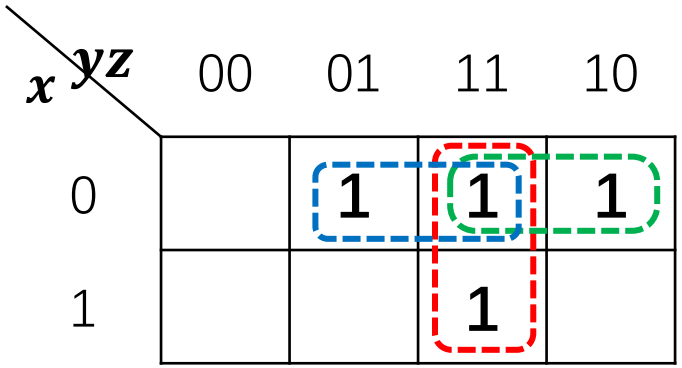


$F = \bar{x}z + x\bar{y} + y\bar{z}$



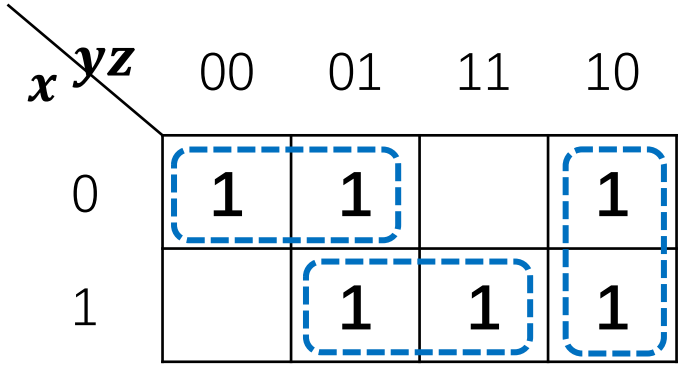
$F = \bar{x}y + x\bar{z} + \bar{y}z$

② $F(x, y, z) = \Sigma m(1, 2, 3, 7)$



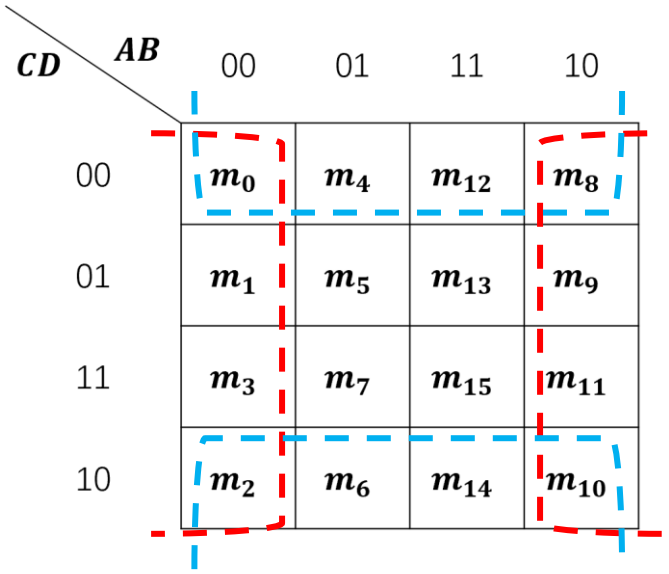
$F = \bar{x}z + yz + \bar{x}y$

③ $F(x, y, z) = \Sigma m(0, 1, 2, 5, 6, 7)$

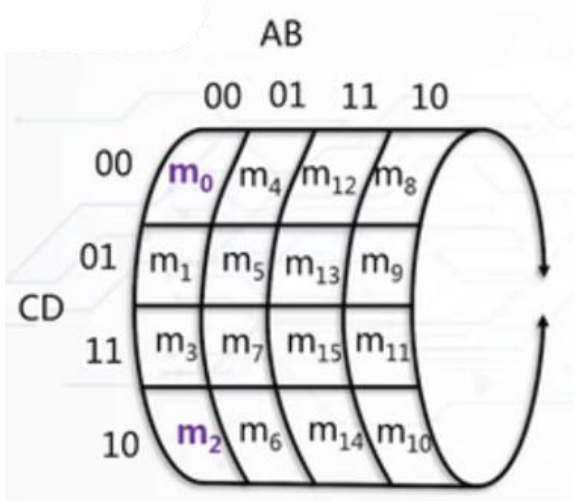


$F = \bar{x}\bar{y} + xz + y\bar{z}$

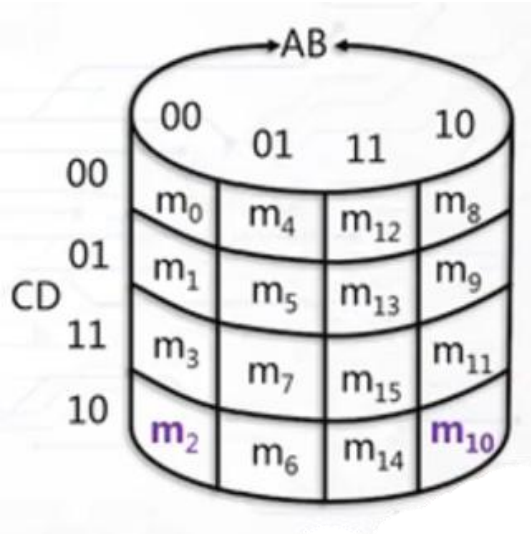
4变量卡诺图

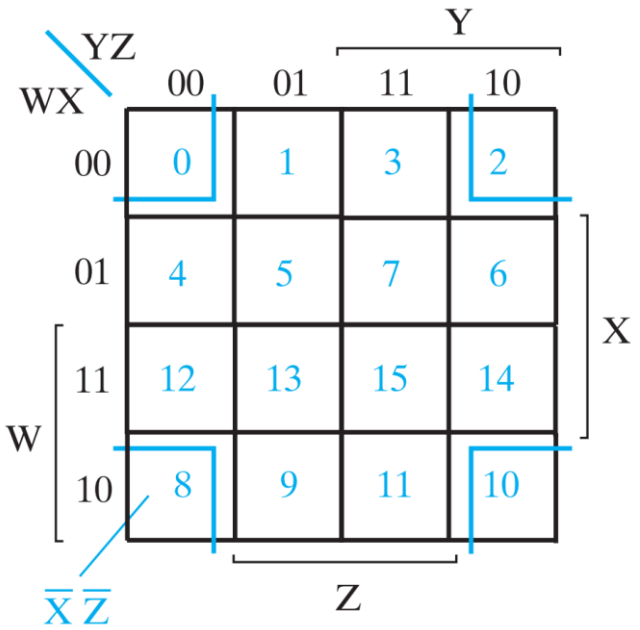


上下两边相邻

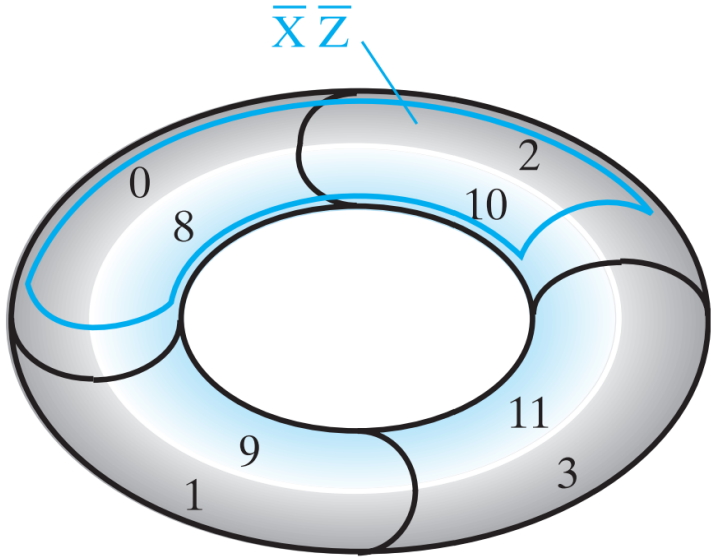


左右两边相邻





4个角是相邻的



用卡诺图化简

AB \ CD	00	01	11	10
00	**1**		**1**	**1**
01		**1**	**1**	
11		**1**	**1**	
10	**1**	**1**	**1**	**1**

本题有4个化简结果:

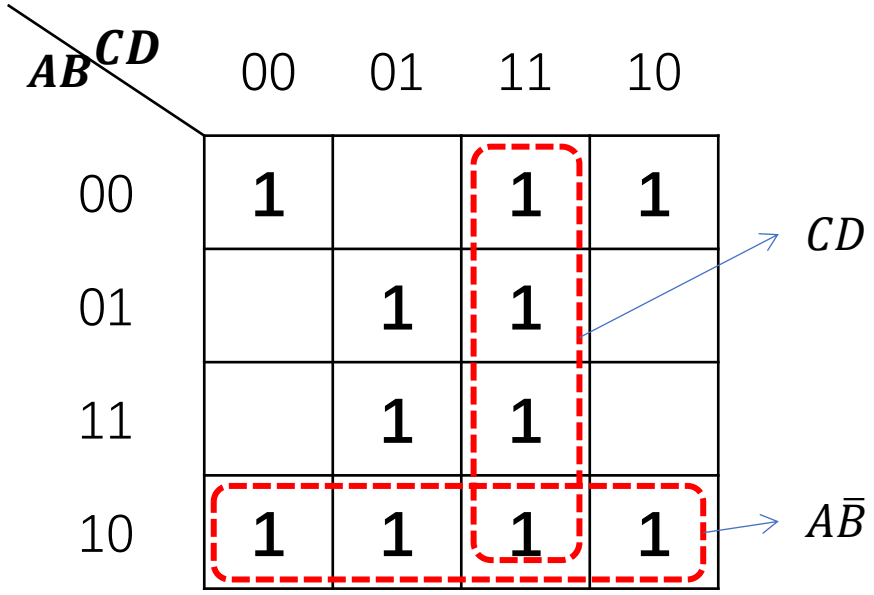
$$F = \boldsymbol{BD} + \overline{\boldsymbol{B}}\overline{\boldsymbol{D}} + AD + CD$$

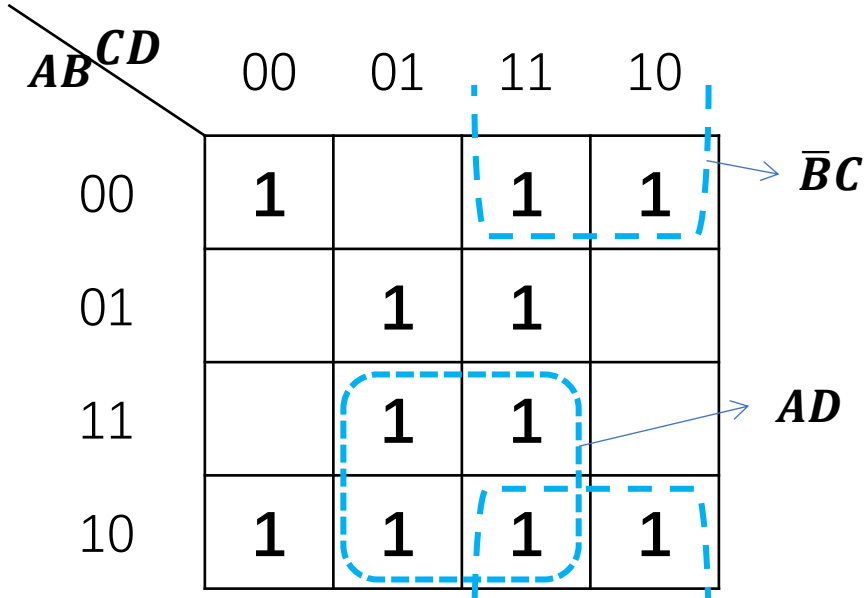
$$F = \boldsymbol{BD} + \overline{\boldsymbol{B}}\overline{\boldsymbol{D}} + A\bar{B} + CD$$

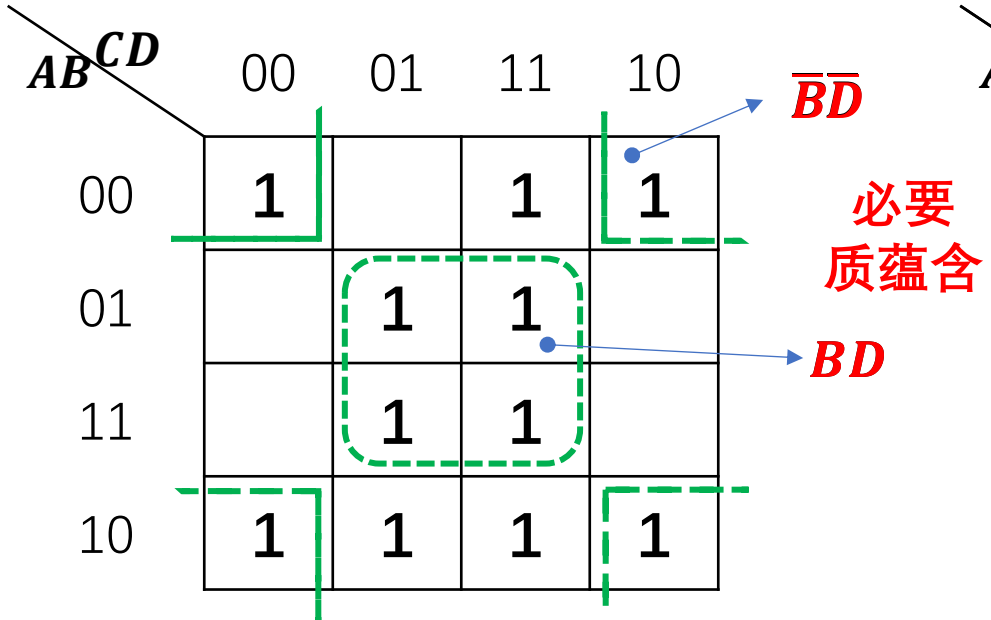
$$F = \boldsymbol{BD} + \overline{\boldsymbol{B}}\overline{\boldsymbol{D}} + AD + \bar{B}C$$

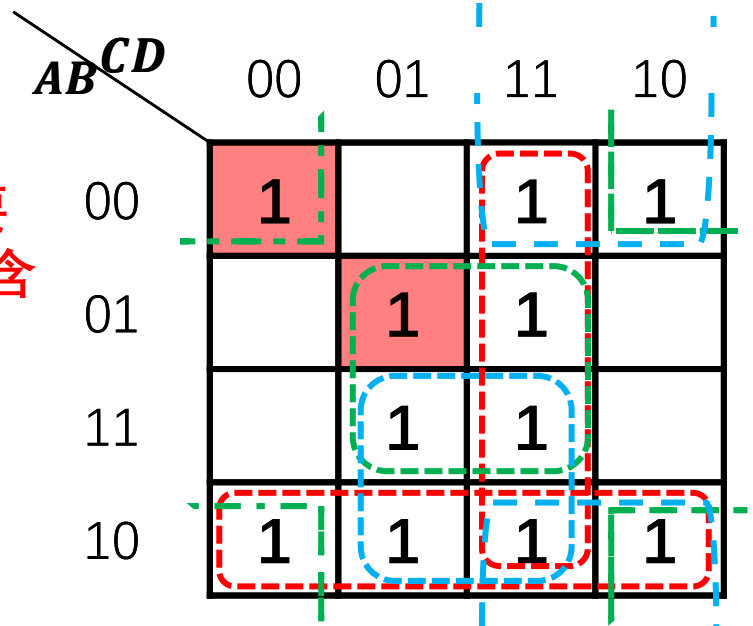
$$F = \boldsymbol{BD} + \overline{\boldsymbol{B}}\overline{\boldsymbol{D}} + A\bar{B} + \bar{B}C$$

①② - 寻找：必要质蕴含



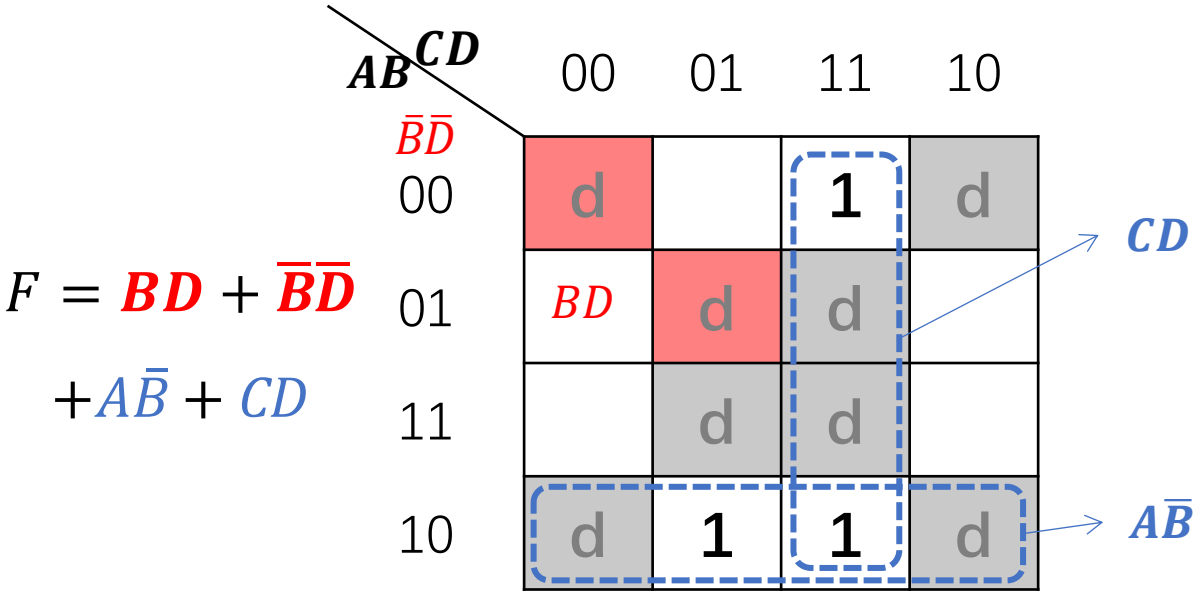


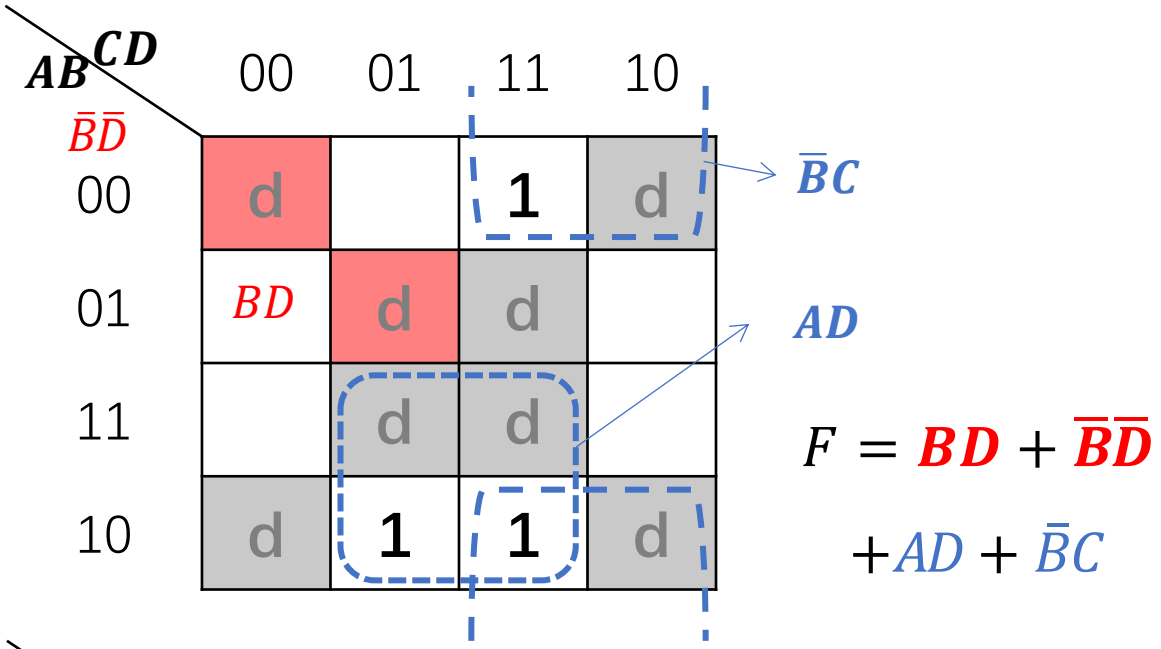


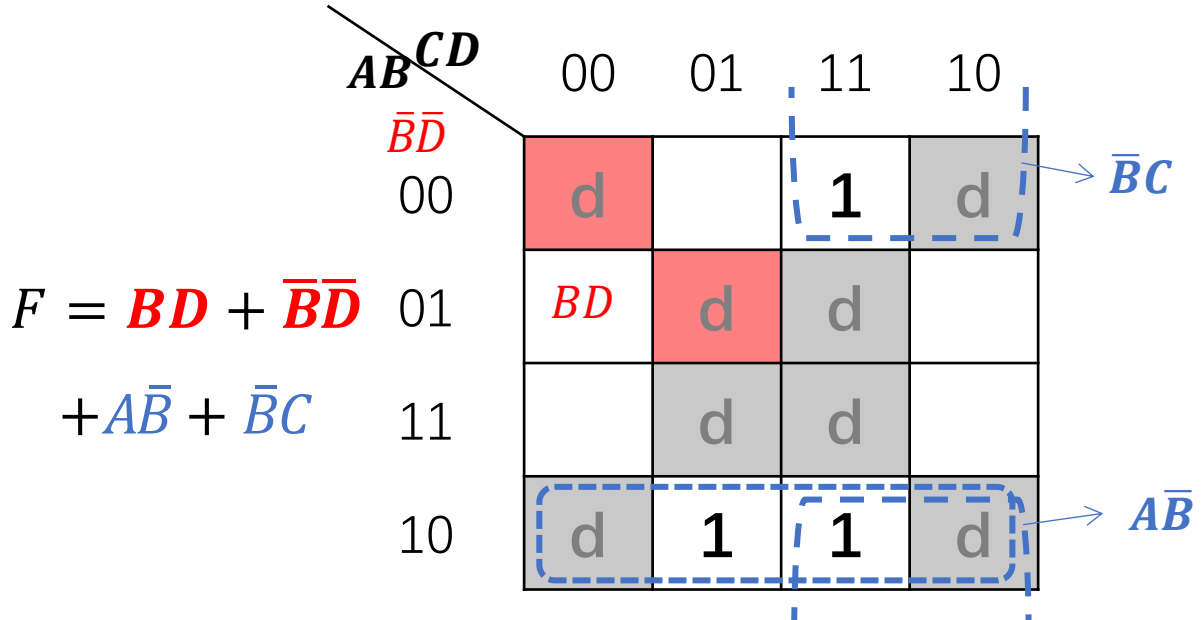


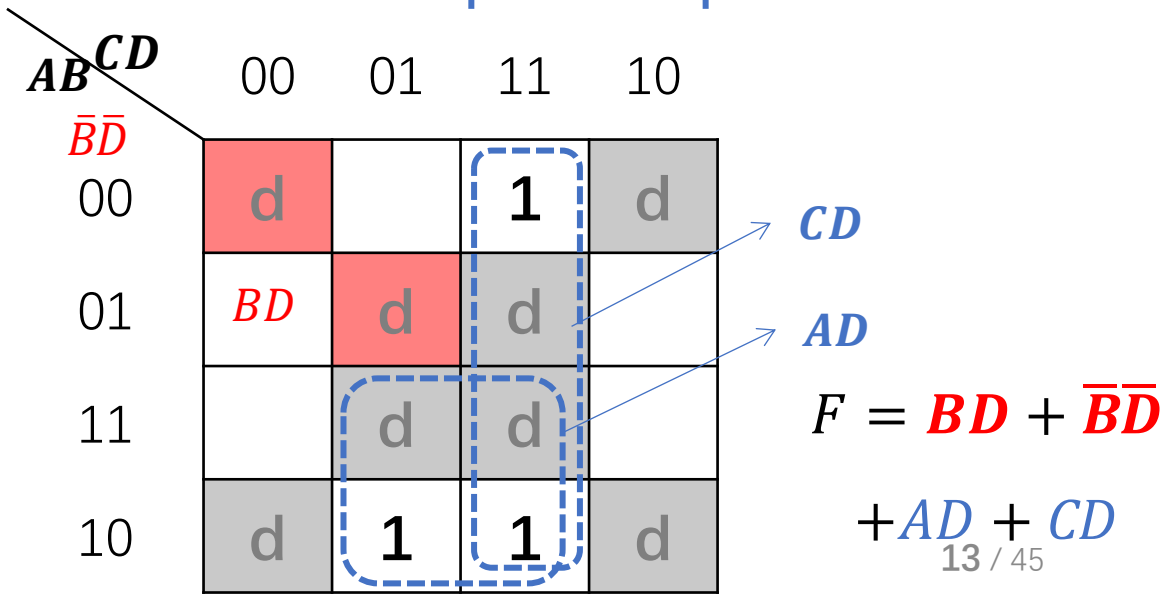
$$F = BD + \bar{B}\bar{D} + \cdots$$

③ 处理其余蕴含（白色底的1）







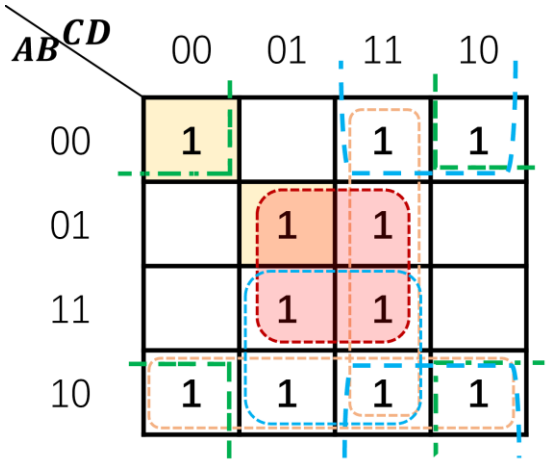


质蕴含 Prime Implicant

- **蕴 含**：函数“**与或**”表达式中的每个”**与**”项。
- 质 蕴 含：卡诺图中**最多**数目的相邻”**与**”项。

  即，该蕴含项不是其他蕴含项的子集。
- **必要质蕴含**：如果**一个最小项**对应的方格只被一个质蕴含所包含，

  那么包含这个最小项的质蕴含，称为**必要质蕴含**。



卡诺图化简方法

① 画出所有的**质蕴含**

② 找出**必要质蕴含**：通过查找一个特殊的乘积项

处理**必要质蕴含**：这是必需要的项

③ 处理**剩余的非必要质蕴含**

【练习3】用卡诺图化简

$$Y = \bar{A}\bar{B}\bar{C}D + \bar{A}B\bar{D} + ACD + A\bar{B}$$

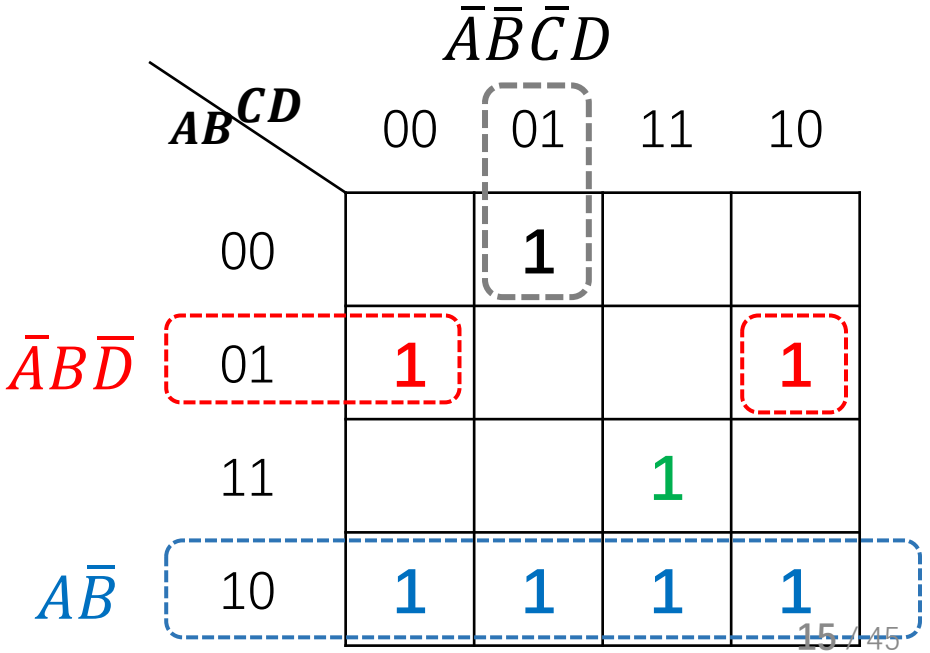
$$= \bar{A}\bar{B}\bar{C}D + \bar{A}B(C + \bar{C})\bar{D} + A(B + \bar{B})CD + A\bar{B}(C + \bar{C})(D + \bar{D})$$

$$= m_1 + m_4 + m_6 + m_8 + m_9 + m_{10} + m_{11} + m_{15}$$

方法1：“**与-或**”表达式转化为**标准**“**与-或**”表达式

AB \ CD	00	01	11	10
00		1		
01	1			1
11			1	
10	1	1	1	1

方法2：直接写出所有的**1**

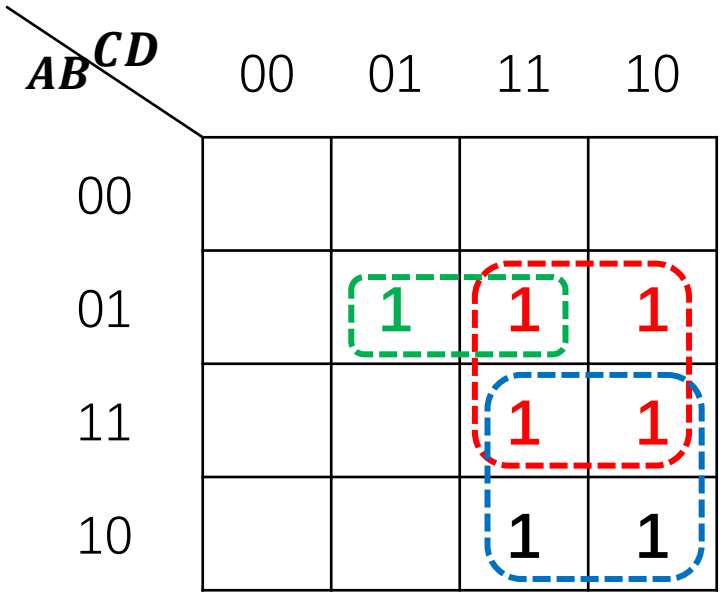


【练习4】用卡诺图化简

① $Y = A\bar{B}C + BC + \bar{A}B\bar{C}D$

化简后:

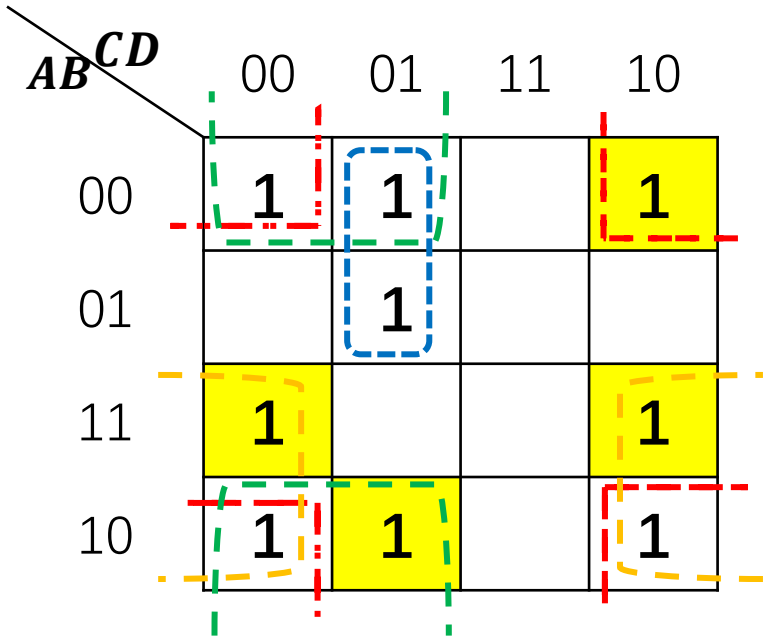
$Y = BC + AC + \bar{A}BD$



② $Y = \Sigma m(0,1,2,5,8,9,10,12,14)$

化简后:

$Y = \bar{B}\bar{C} + \bar{B}\bar{D} + A\bar{D} + \bar{A}\bar{C}D$



【证明】$AB + \bar{A}C + (\bar{B} + \bar{C})D = \boldsymbol{AB + \bar{A}C + D}$

如果公式推导有困难，可以借助卡诺图帮助。

有多种方案，这是其中一种：寻找$\boldsymbol{D}$。

证： $AB + \bar{A}C + (\bar{B} + \bar{C})\boldsymbol{D}$

①　②　③　④

$= \boldsymbol{AB} + \boldsymbol{\bar{A}C} + \boldsymbol{\bar{B}D} + \boldsymbol{\bar{C}D}$

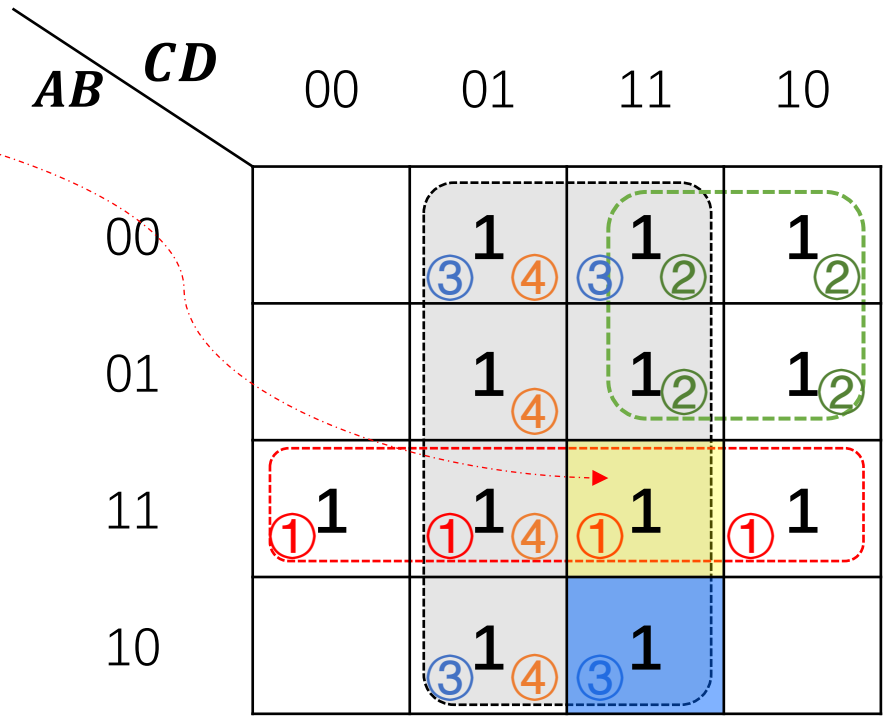
$= AB + \boldsymbol{ABCD} + \bar{A}C + \bar{B}D + \boldsymbol{A\bar{B}CD} + \bar{C}D$

$= AB + \bar{A}C + \boldsymbol{\bar{A}CD} + \bar{B}D + \bar{C}D + \boldsymbol{ACD}$

$= AB + \bar{A}C + \bar{B}D + \bar{C}D + \boldsymbol{CD}$

$= AB + \bar{A}C + \bar{B}D + \boldsymbol{D}$

$= AB + \bar{A}C + D$



【证明】 $\overline{AB + \bar{A}C} = A\bar{B} + \bar{A}\bar{C}$

$$\overline{\boldsymbol{AB} + \bar{\boldsymbol{A}}\boldsymbol{C}} = \overline{\boldsymbol{AB}} \cdot \overline{\bar{\boldsymbol{A}}\boldsymbol{C}}$$

$$= (\bar{A} + \bar{B}) \cdot (A + \bar{C})$$

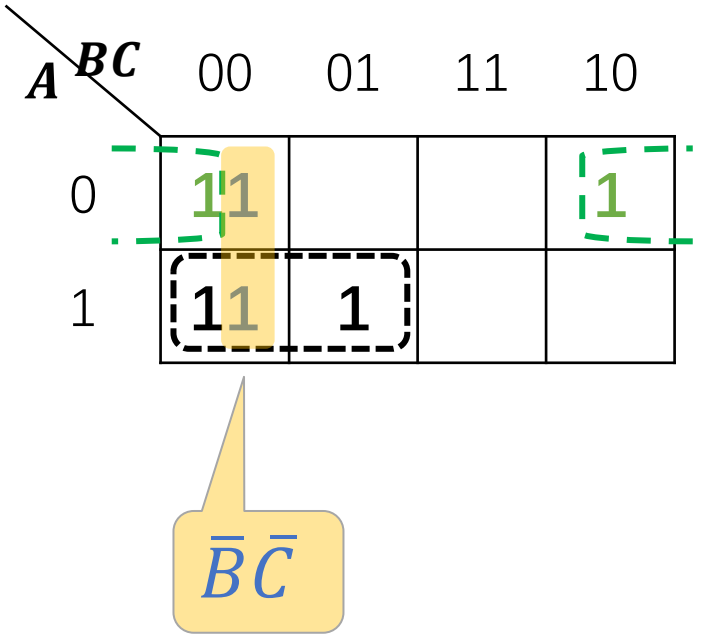
$$= \bar{A}A + \bar{A}\bar{C} + A\bar{B} + \bar{B}\bar{C}$$

$$= \bar{A}\bar{C} + A\bar{B} + \bar{B}\bar{C}(A + \bar{A})$$

$$= \bar{A}\bar{C} + A\bar{B} + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}$$

$$= \bar{A}\bar{C}(1 + \bar{B}) + A\bar{B}(1 + \bar{C})$$

$$= \bar{A}\bar{C} + A\bar{B}$$



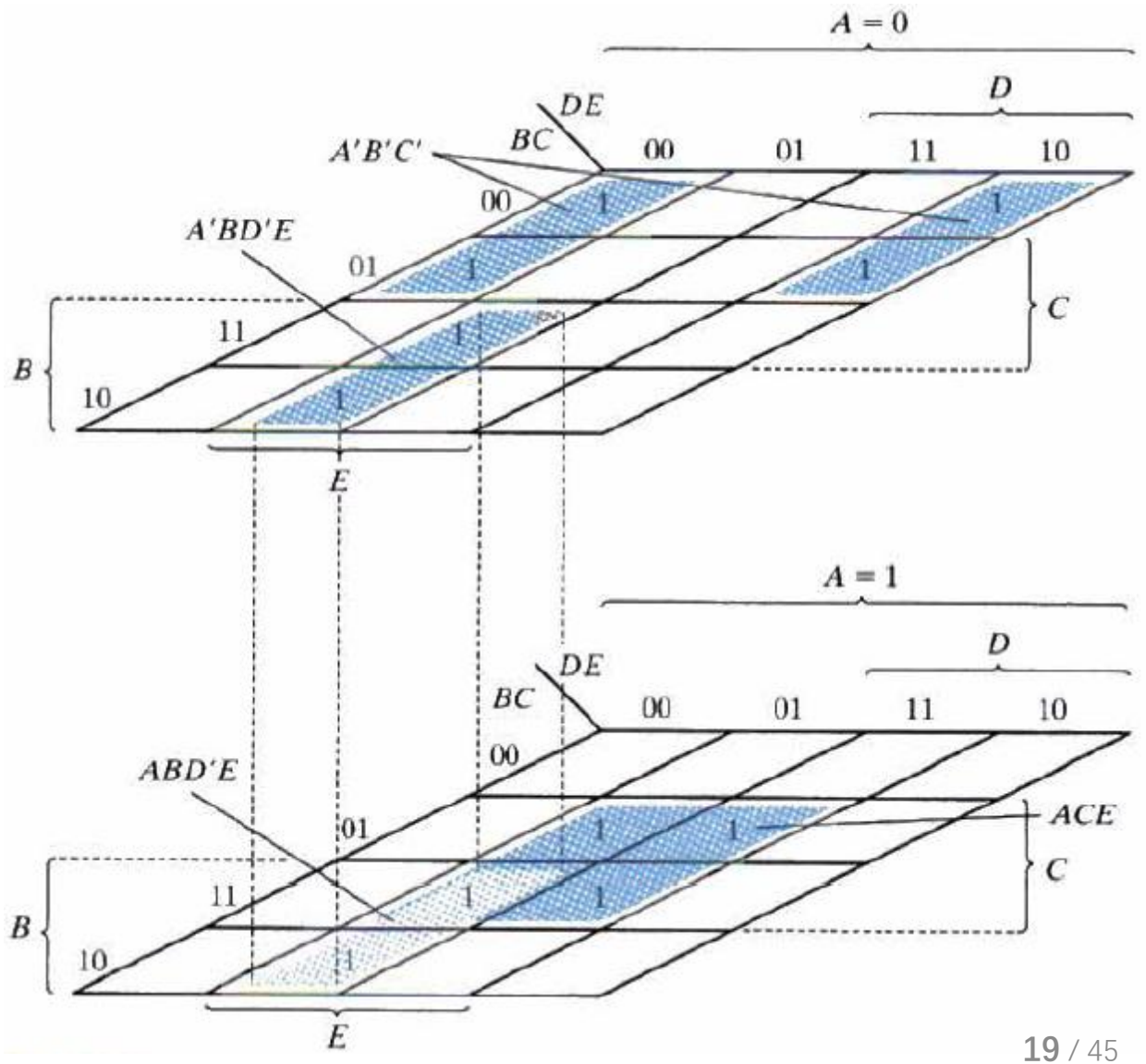
5输入变量卡诺图

A=0

BC \ DE	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

A=1

BC \ DE	00	01	11	10
00	16	17	19	18
01	20	21	23	22
11	28	29	31	30
10	24	25	27	26

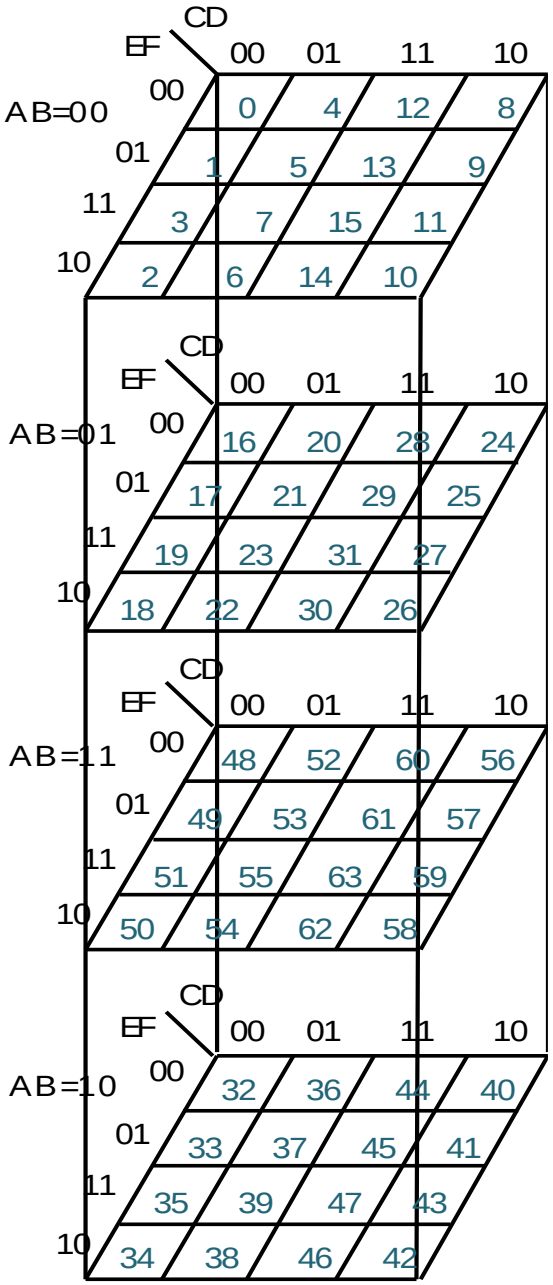


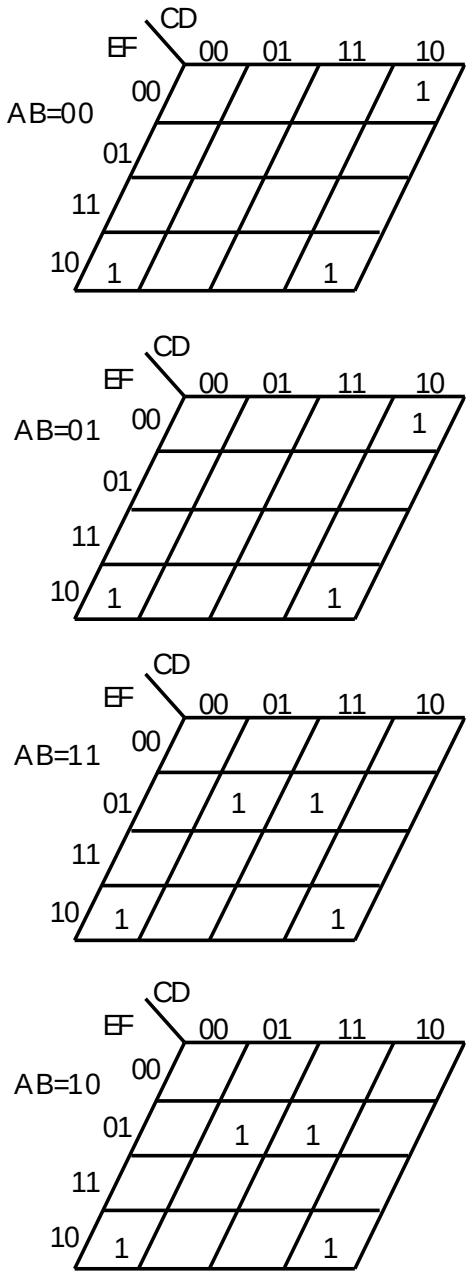
6输入变量卡诺图

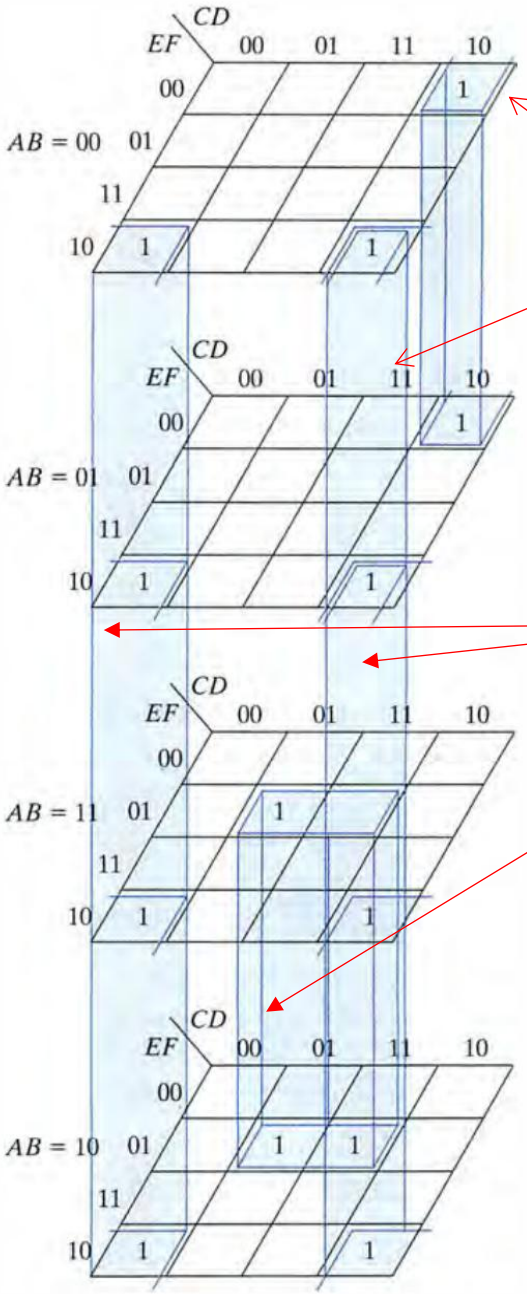
$$F(A,B,C,D,E,F) = \Sigma m\ (2,8,10,18,24,26,34,37,42,45,50,53,58,61)$$

ABC \ DEF	000	001	011	010	100	101	111	110
000				2				
001	8			10				
011	24			26				
010				18				
100				34		37		
101				42		45		
111				58		61		
110				50		53		

6输入变量卡诺图







$$F = \bar{D}E\bar{F} + AD\bar{E}F + \bar{A}C\bar{D}\bar{F}$$

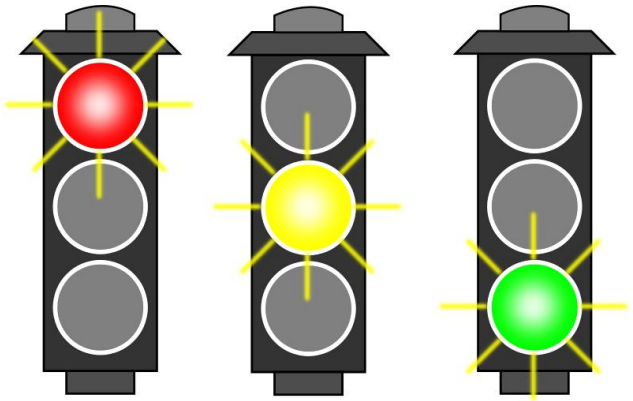
2

无关项

完全描述、非完全描述

- **完全描述逻辑函数**：对于输入变量的每一组取值，逻辑函数都有确定的值。
- **非完全描述逻辑函数**：对于输入变量的某些取值组合，逻辑函数值不确定($\boldsymbol{d}$)，

即函数值可以为0，也可以为1，而不影响函数的功能。如，红绿交通灯。



假设：1-灯亮、0-灯灭

1-通行、0-停止

红灯 R	黄灯 Y	绿灯 G	通行 F
0	0	0	d
0	0	1	1
0	1	0	0
0	1	1	d
1	0	0	0
1	0	1	d
1	1	0	d
1	1	1	d

$$F = \sum m(1) + \sum d(0,3,5,6,7)$$

R \ YG	00	01	11	10
0	d	1	d	0
1	0	d	d	d

$$F = G$$

无关项 (don’t-care)

- 实际应用中，函数不是由确定的0或1变量组合规定的。
- 大多数情况下，我们不关心这些不确定项给函数值带来的影响。
- 这些不确定项 ($\boldsymbol{d}$) 或 (X)称为：**无关项**。
- 无关项可以用于卡诺图的**进一步化简**。

化简 $Y(A,B,C,D)=\sum m(1,7,8)+\boldsymbol{d}(3,5,9,10,12,14,15)$

也可以写作：　　$Y=\bar{A}\bar{B}\bar{C}D+\bar{A}BCD+A\bar{B}\bar{C}\bar{D}$

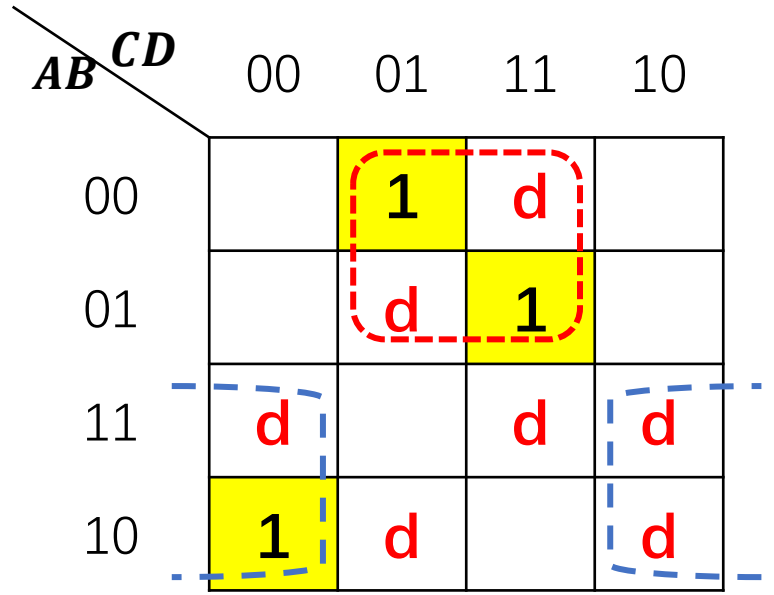
$$\bar{A}\bar{B}CD+\bar{A}B\bar{C}D+AB\bar{C}\bar{D}+A\bar{B}\bar{C}D+ABCD+ABC\bar{D}+A\bar{B}C\bar{D}=0$$

【解】这个逻辑函数已无可化简，但适当加入约束后还可以化简

$$Y=(\bar{A}\bar{B}\bar{C}D+\bar{A}\bar{B}CD)+(\bar{A}BCD+\bar{A}B\bar{C}D)+(A\bar{B}\bar{C}\bar{D}+AB\bar{C}\bar{D})+(ABC\bar{D}+A\bar{B}C\bar{D})$$

$$=(\bar{A}\bar{B}D+\bar{A}BD)+(A\bar{C}\bar{D}+AC\bar{D})$$

$$=\boldsymbol{\bar{A}D+A\bar{D}}$$



$F\ (w,x,y,z) = \Sigma m(1,3,7,11,15)\ ,\quad \boldsymbol{d}(w,x,y,z) = \Sigma m(0,2,5)$

wx \ yz	00	01	11	10
00		1	1	
01			1	
11			1	
10			1	

$F = yz + \bar{w}\bar{x}z$

wx \ yz	00	01	11	10
00	d	1	1	d
01		d	1	
11			1	
10			1	

$F = yz + \bar{w}\bar{x}$

无关项简化了逻辑函数表达式。

wx \ yz	00	01	11	10
00	1	1	1	1
01		1	1	
11			1	
10			1	

如果无关项处均为1

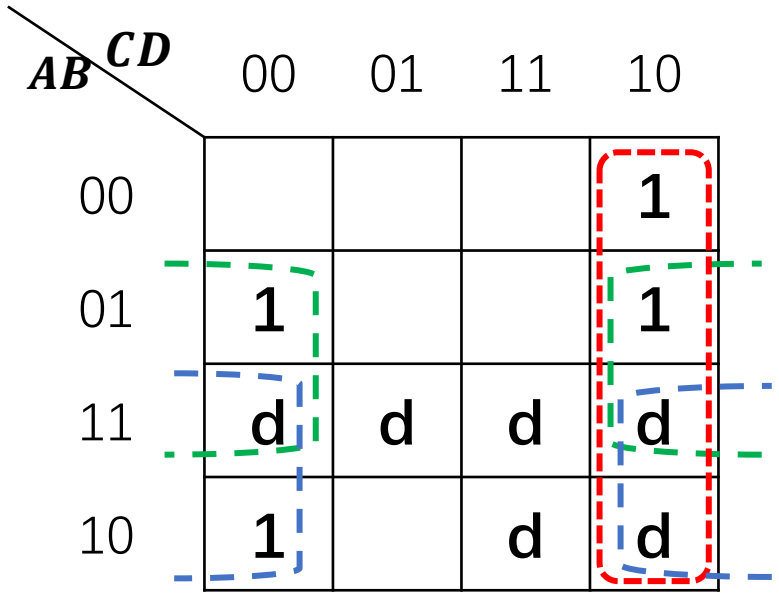
$F = yz + \bar{w}z + \bar{w}\bar{x}$

wx \ yz	00	01	11	10
00	d	1	1	d
01		d	1	
11			1	
10			1	

$F = yz + \bar{w}z$

【练习7】化简

$$Y(A,B,C,D)=\sum m(2,4,6,8)+d(10,11,12,13,14,15)$$



$$Y=B\overline{D}+A\overline{D}+C\overline{D}$$

3

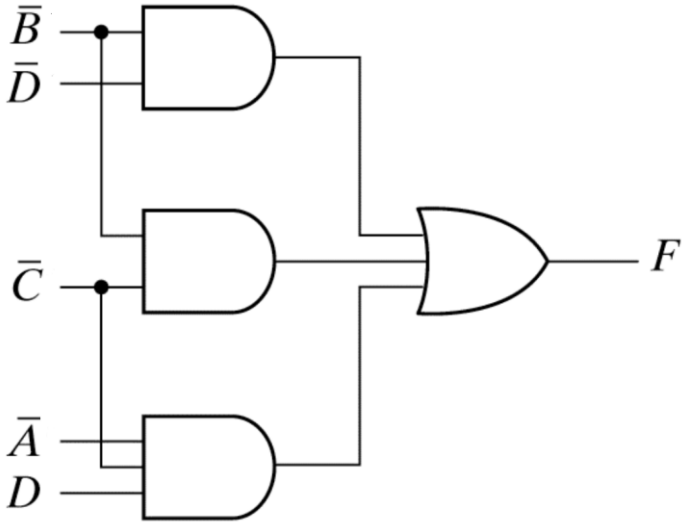
其它形式

2种形式的二级门电路

标准式1：积之和

① 在真值表上找 $F=1$的项

② $F=\bar{B}\bar{D}+\bar{B}\bar{C}+\bar{A}\bar{C}D$



标准式2：和之积

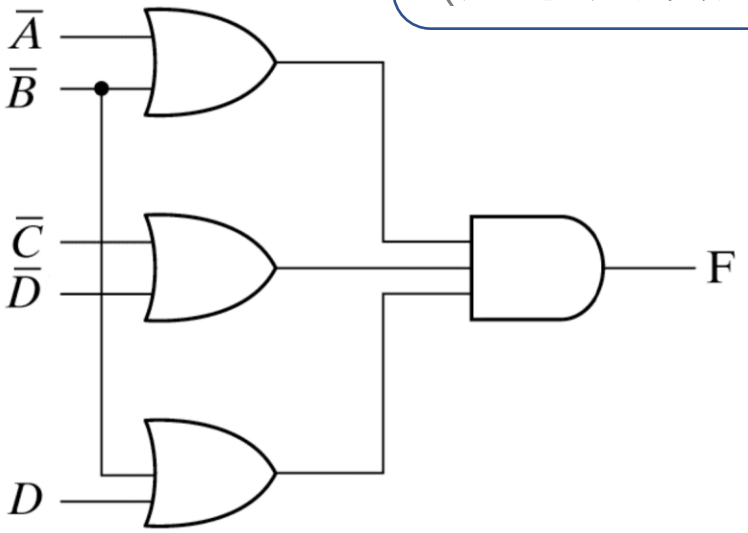
① 在真值表上找 $F=\mathbf{0}$的项，即 $\bar{F}$

② $\bar{F}=AB+CD+B\bar{D}$

③ $F=\bar{\bar{F}}=\overline{AB+CD+B\bar{D}}$ (与或非式)

$=(\bar{A}+\bar{B})(\bar{C}+\bar{D})(\bar{B}+D)$

(用**最大项**法可以直接化简得到)



AB \ CD	00	01	11	10
00	1	1	0	1
01	0	1	0	0
11	0	0	0	0
10	1	1	0	1

最大项法 $F(w,x,y,z)=\Sigma m(1,3,7,11,15),\ \boldsymbol{d}=\Sigma m(0,2,5)$

wx \ yz	00	01	11	10
00	0	1	1	0
01	0	0	1	0
11	0	0	1	0
10	0	0	1	0

d 都为0时：

$F=z(\bar{w}+y)(\bar{x}+y)$

d 都为1时：

$F=(\bar{w}+y)(\bar{w}+z)(\bar{x}+z)$

wx \ yz	00	01	11	10
00	1	1	1	1
01	0	1	1	0
11	0	0	1	0
10	0	0	1	0

wx \ yz	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	0	1	0
10	0	0	1	0

$F=z(\bar{w}+y)$

无关项简化了逻辑函数表达式。

用两次取反法 求 最简“或-与”表达式

$$F = A\bar{C} + AD + \bar{B}\bar{C} + \bar{B}D$$

卡诺图合并所有0的最小项，再公式取反

CD \ AB	00	01	11	10
00	1	0	1	1
01	1	0	1	1
11	1	0	1	1
10	0	0	0	0

- F的卡诺图
- $\overline{F}$的化简
合并0项
- $\overline{F}$ 取反

$$\overline{F} = \bar{A}B + C\bar{D}$$

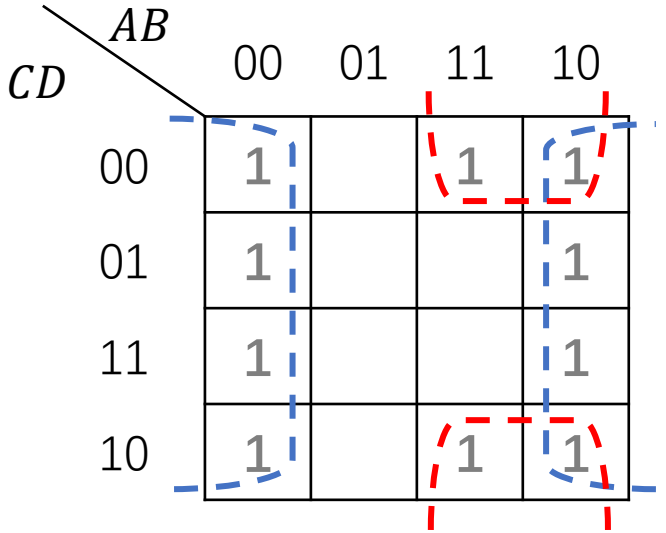
$$F = \bar{\bar{F}} = (A + \bar{B})(\bar{C} + D)$$

$$F = (\bar{A} + D)(B + \bar{D})(A + B)$$

公式取反，卡诺图化简，公式再取反

$$\overline{F} = A\bar{D} + \bar{B}D + \bar{A}\bar{B}$$

- $\overline{F}$的表达式



- $\overline{F}$的卡诺图
- $\overline{F}$的化简
合并1项
- $\overline{F}$ 取反

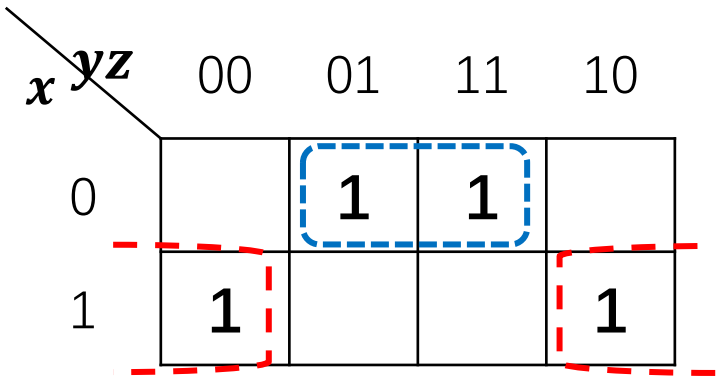
$$\overline{F} = \bar{B} + A\bar{D}$$

$$F = \bar{\bar{F}} = B\,(\bar{A} + D)$$

【练习5】化简 $F(x,y,z)=\sum(1,3,4,6)$

① 积之和的形式

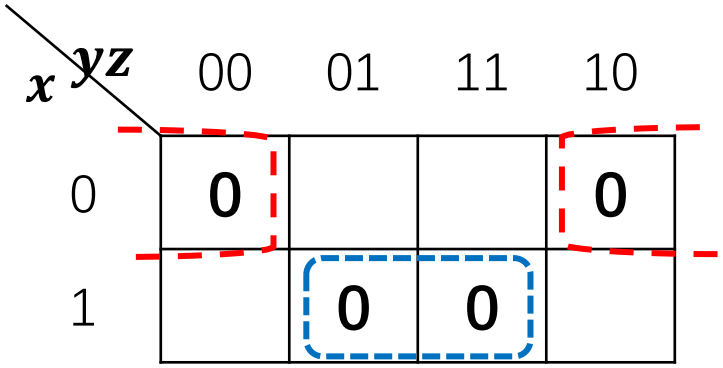
$$F(x,y,z)=\sum m(1,3,4,6)$$



$$F=\overline{x}z+x\overline{z}$$

② 和之积的形式

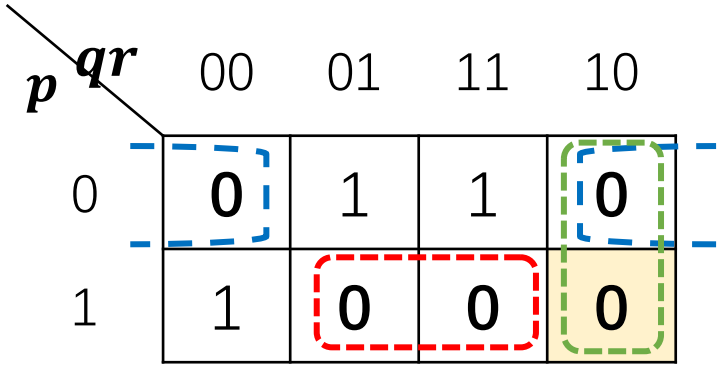
$$F(x,y,z)=\prod M(0,2,5,7)$$



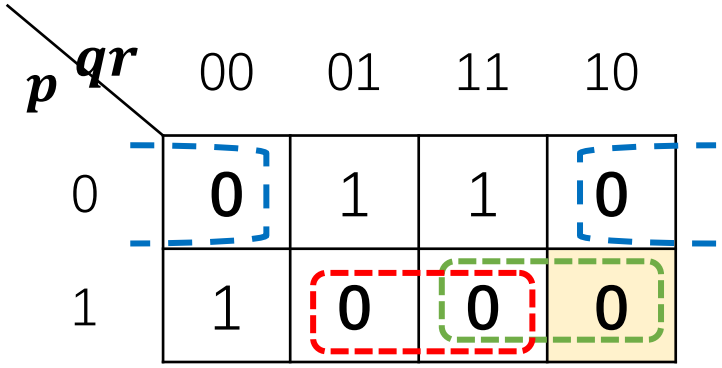
$$\overline{F}=xz+\overline{x}\overline{z}$$

$$F=(\overline{x}+\overline{z})(x+z)$$

【练习6】化简 $(p+r)(\bar{p}+\bar{q})(\bar{q}+r)(\bar{p}+\bar{r})$



$$(p+r)(\bar{q}+r)\ (\bar{p}+\bar{r})$$



$$(p+r)(\bar{p}+\bar{q})(\bar{p}+\bar{r})$$

奇函数

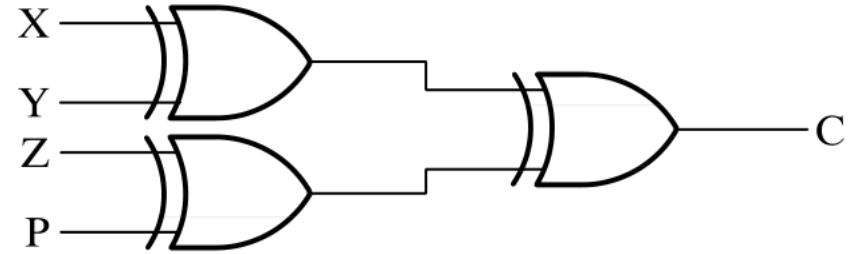
只有奇数个输入变量＝1时，输出值才＝1

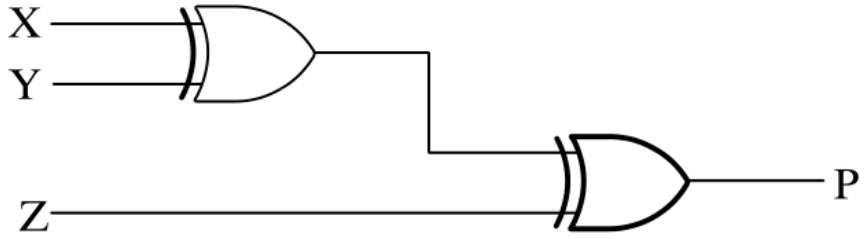
$$X \oplus Y = X\overline{Y} + \overline{X}Y$$

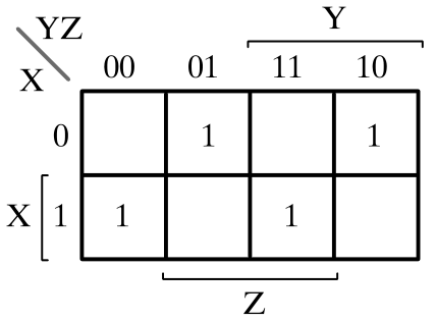
相异为1

$$X \oplus Y \oplus Z = (X\overline{Y} + \overline{X}Y)\overline{Z} + (XY + \overline{X}\,\overline{Y})Z$$

$$= X\overline{Y}\,\overline{Z} + \overline{X}Y\overline{Z} + \overline{X}\,\overline{Y}Z + XYZ$$

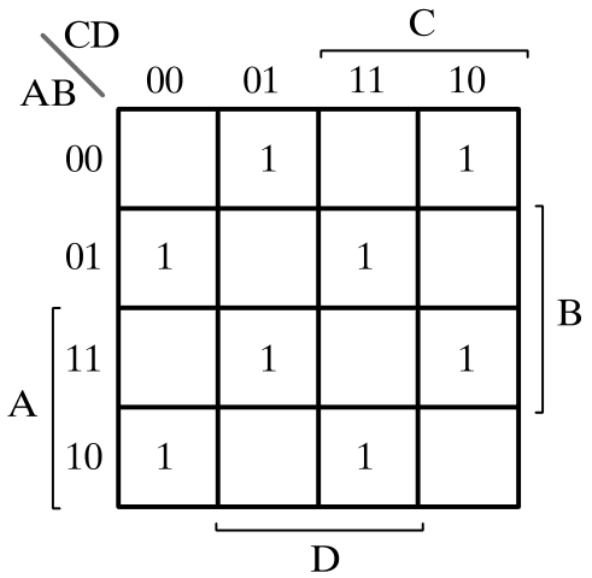






(a) X ⊕ Y ⊕ Z

三变量或三变量以上的⊕运算都是奇函数。



(b) A ⊕ B ⊕ C ⊕ D

教材P56，2.1e)

$$Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + AB\bar{C}\bar{D} + ABCD$$

(e)

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

AB \ CD	00	01	11	10
00	**1**		**1**	
01		**1**		**1**
11	**1**		**1**	
10		**1**		**1**

$$Y = \overline{(A \oplus B)}\,\overline{(C \oplus D)} + (A \oplus B)(C \oplus D)$$

反函数、对偶式

$F = \bar{x}z + x\bar{y}$

x \ yz	00	01	11	10
0		1	1	
1	1	1		

反函数 $\bar{F}$

x \ yz	00	01	11	10
0	1	0	0	1
1	0	0	1	1

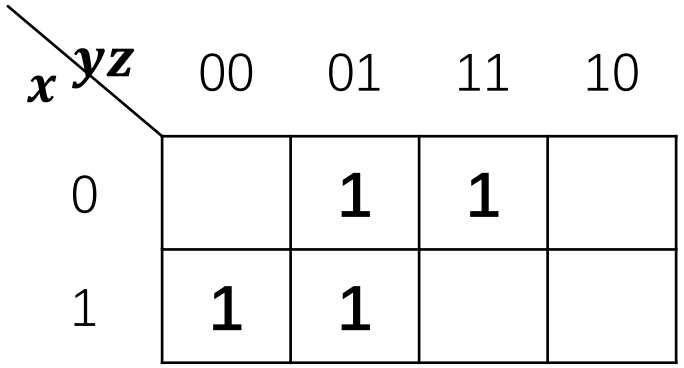
对偶 $F' = (\bar{x} + z)(x + \bar{y})$

$= \bar{x}\bar{y} + xz + \bar{y}z$

x \ yz	00	01	11	10
0	1	1		
1		1	1	

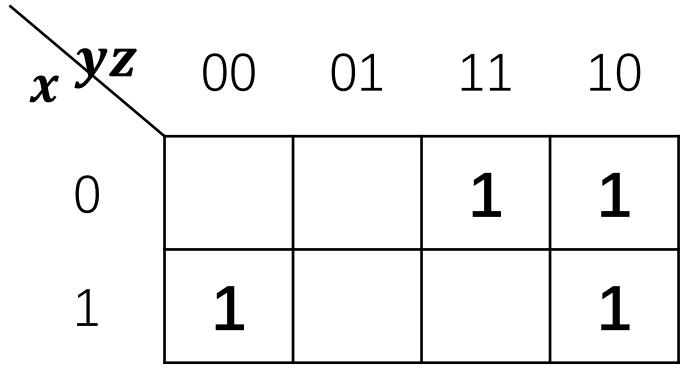
用卡诺图做运算

$F1$



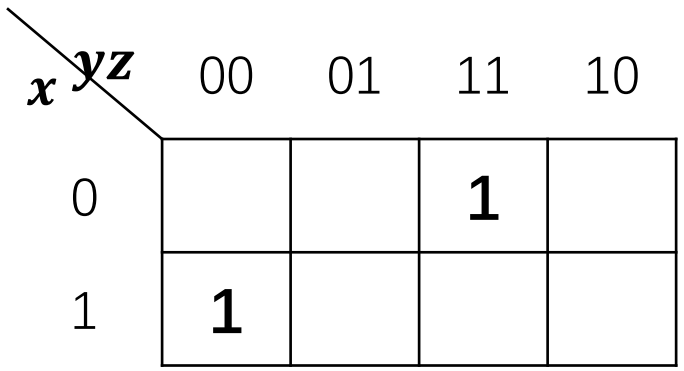
x \ yz	00	01	11	10
0		1	1	
1	1	1		

$F2$



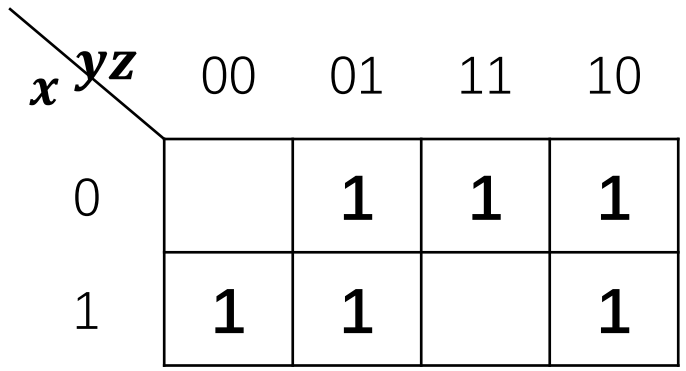
x \ yz	00	01	11	10
0			1	1
1	1			1

$F1 \cdot F2$



x \ yz	00	01	11	10
0			1	
1	1			

$F1 + F2$



x \ yz	00	01	11	10
0		1	1	1
1	1	1		1

用卡诺图做运算(含无关项)

$F1$

x \ yz	00	01	11	10
0		**1**	d	
1	**1**	**1**	d	

$F2$

x \ yz	00	01	11	10
0			**1**	**1**
1	**1**			**1**

$0+d=d \quad 1+d=1 \quad 0\cdot d=0 \quad 1\cdot d=d \quad d+d=d \quad d\cdot d=d \quad \bar{d}=d$

$F1\cdot F2$

x \ yz	00	01	11	10
0			d	
1	**1**			

$F1+F2$

x \ yz	00	01	11	10
0		**1**	**1**	**1**
1	**1**	**1**	d	**1**

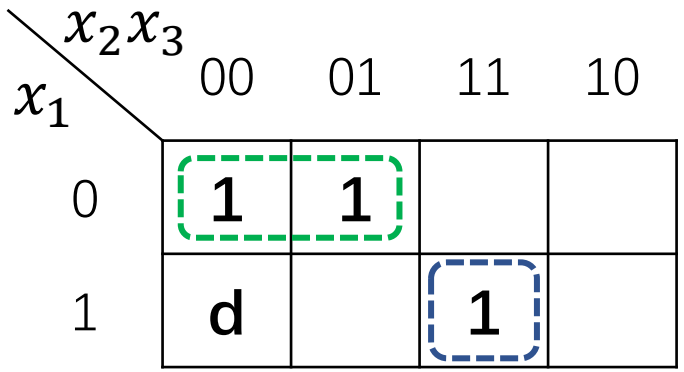
4

多输出

【练习8】多输出函数设计最优化

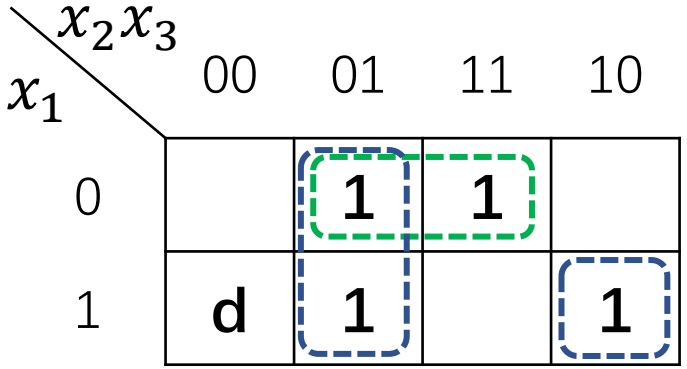
$x_1x_2x_3$	y_1y_2
0 0 0	1 0
0 0 1	1 1
0 1 0	0 0
0 1 1	0 1
1 0 0	d d
1 0 1	0 1
1 1 0	0 1
1 1 1	1 0

y_1



$$y_1 = \bar{x}_1\bar{x}_2 + x_1x_2x_3$$

y_2

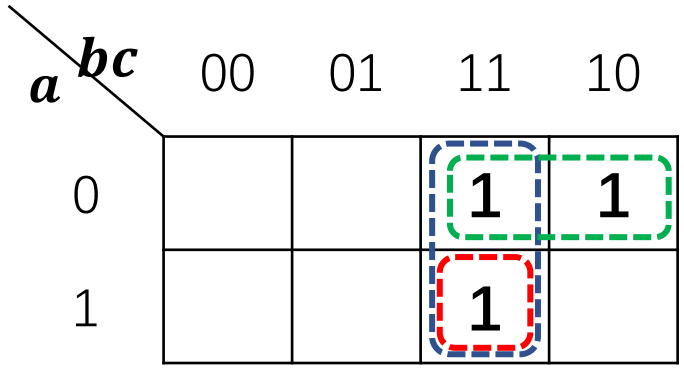


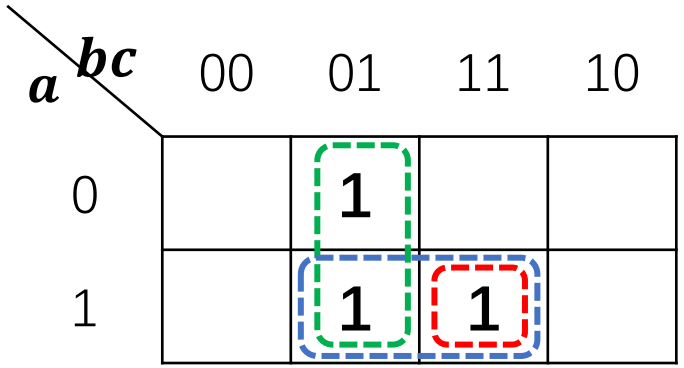
$$y_2 = \bar{x}_1x_3 + \bar{x}_2x_3 + x_1x_2\bar{x}_3$$

- **单输出函数**化简只须符合要求的逻辑形式达到最小覆盖即可。但
- **多输出函数**由于存在各个函数之间的相关性，需要考虑**公共项**!

【例】化简$F_1(a,b,c)=\Sigma m(2,3,7)$;

$$F_2(a,b,c)=\Sigma m(1,5,7)$$





公共最小项是m_7

$F_1=\bar{a}b+bc$

$F_1=\bar{a}b+abc$

总乘积项为 4 个

总乘积项为 3 个

$F_2=\bar{b}c+ac$

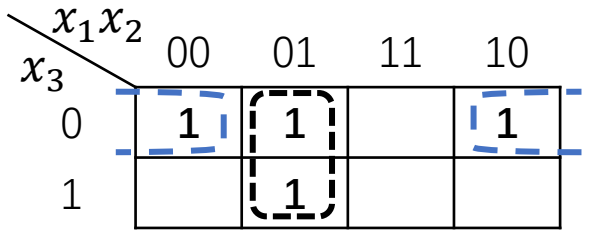
$F_2=\bar{b}c+abc$

质蕴涵		m_0	m_1	m_2	m_3	m_4	m_5	m_6	m_7
F_1	$\bar{a}b$			V	V				
	bc				V				V
F_2	$\bar{b}c$		V				V		
	ac						V		V

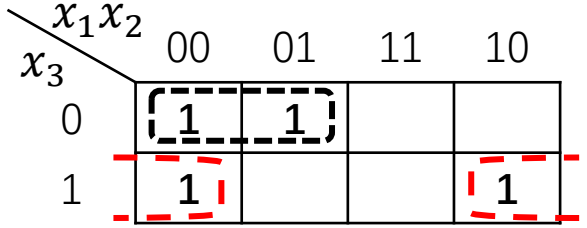
【例】设计最优化

$x_1x_2x_3$	$f^1f^2f^3$
0 0 0	1 1 0
0 0 1	0 1 d
0 1 0	1 1 1
0 1 1	1 0 1
1 0 0	1 0 1
1 0 1	0 1 1
1 1 0	0 0 1
1 1 1	0 0 0

f^1

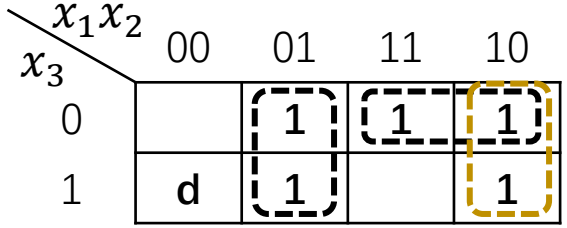


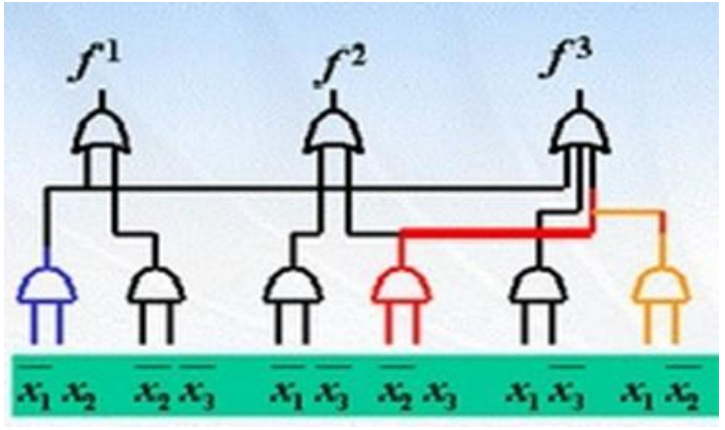
f^2

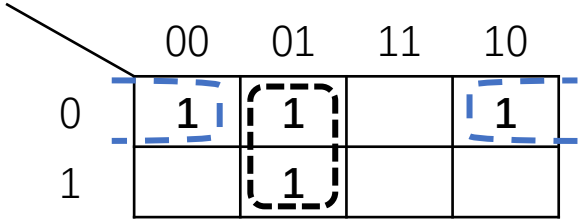


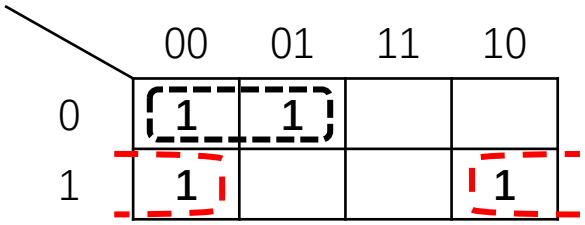
方案1

f^3

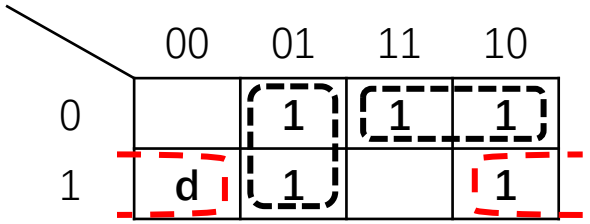


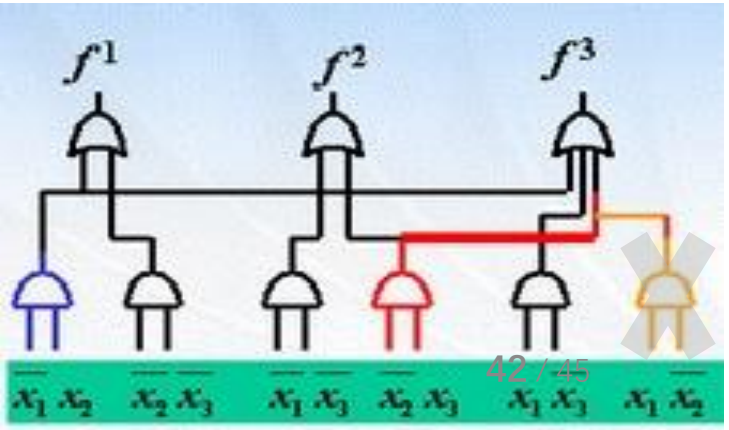






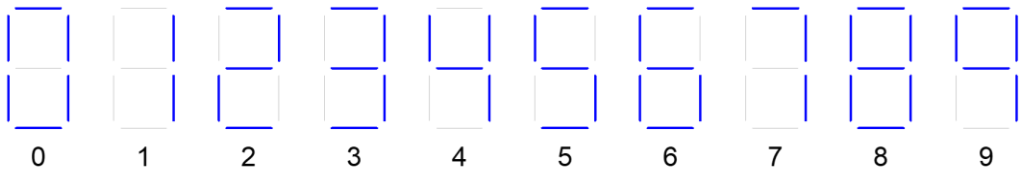
方案2

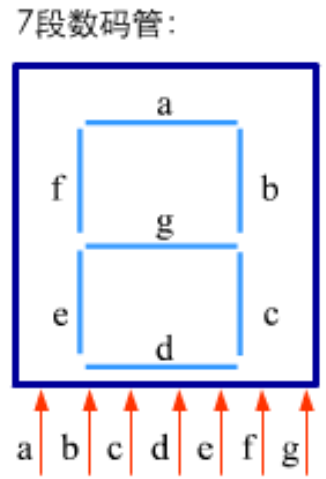


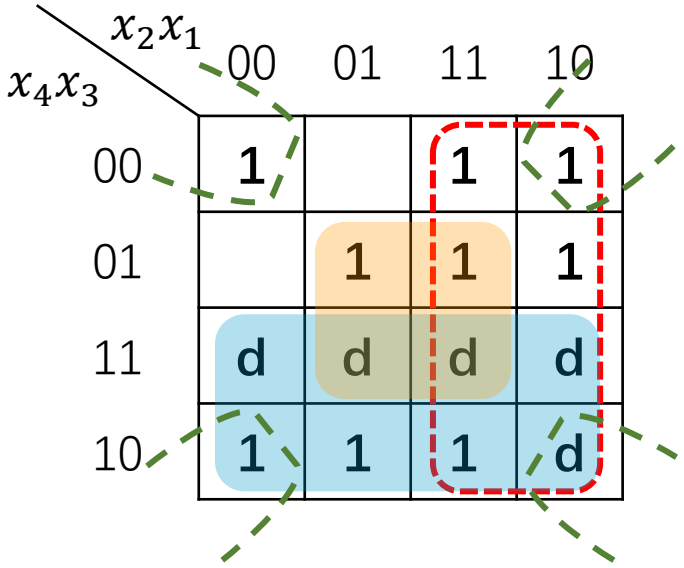


七段数码管BCD设计

	$x_4x_3x_2x_1$	$a\ b\ c\ d\ e\ f\ g$
0	0 0 0 0	1 1 1 1 1 1 0
1	0 0 0 1	0 1 1 0 0 0 0
2	0 0 1 0	1 1 0 1 1 0 1
3	0 0 1 1	1 1 1 1 0 0 1
4	0 1 0 0	0 1 1 0 0 1 1
5	0 1 0 1	1 0 1 1 0 1 1
6	0 1 1 0	1 0 1 1 1 1 1
7	0 1 1 1	1 1 1 0 0 0 0
8	1 0 0 0	1 1 1 1 1 1 1
9	1 0 0 1	1 1 1 1 0 1 1
	1 0 1 0	$d\ d\ d\ d\ d\ d\ d$
	……	……
	1 1 1 1	$d\ d\ d\ d\ d\ d\ d$

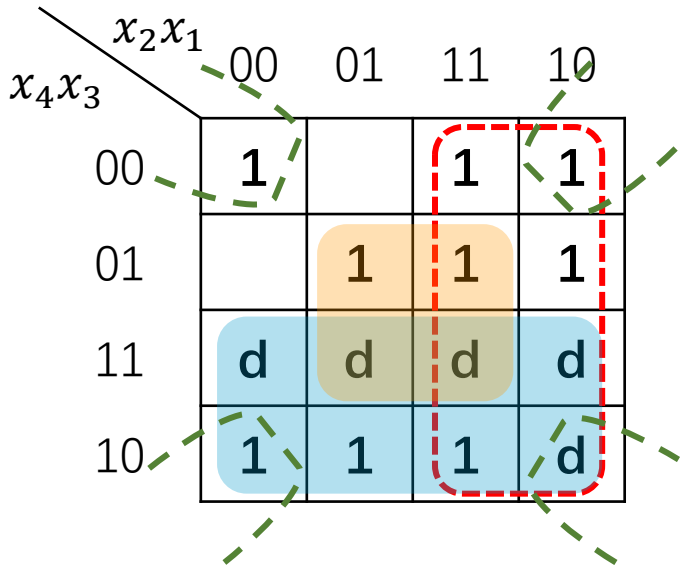


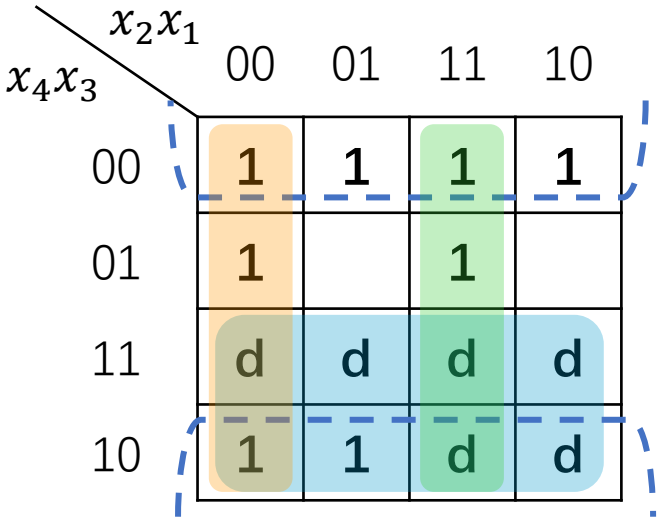




$$a = x_2 + x_3x_1 + x_4 + \overline{x}_3\overline{x}_1$$

七段数码管BCD设计-2





单输出优化

$$\boldsymbol{a} = x_4 + x_3x_1 + x_2 + \bar{x}_3\bar{x}_1$$

$$\boldsymbol{b} = x_4 + \bar{x}_3 + x_2x_1 + \bar{x}_2\bar{x}_1$$

$$c = x_4 + x_3 + \bar{x}_2 + x_1$$

$$d = \bar{x}_3\bar{x}_1 + x_2\bar{x}_1 + x_3\bar{x}_2x_1 + \bar{x}_3x_2$$

$$e = \bar{x}_3\bar{x}_1 + x_2\bar{x}_1$$

$$f = x_4 + \bar{x}_2\bar{x}_1 + x_3\bar{x}_1 + x_3\bar{x}_2$$

$$g = x_4 + \bar{x}_2x_1 + \bar{x}_3x_2 + x_3\bar{x}_1$$

如何同时兼顾7张卡诺图？

七段数码管BCD设计-3

$$a = x_4 + x_3x_1 + x_2 + \bar{x}_3\bar{x}_1$$

$$b = x_4 + \bar{x}_3 + x_2x_1 + \bar{x}_2\bar{x}_1$$

$$a = x_4 + x_2x_1 + \bar{x}_3\bar{x}_1 + x_3\bar{x}_2x_1 + x_3x_2\bar{x}_1$$

$$b = \bar{x}_3x_1 + \bar{x}_3\bar{x}_1 + x_2x_1 + \bar{x}_2\bar{x}_1$$

$$c = \bar{x}_3x_1 + x_3\bar{x}_2x_1 + \bar{x}_2\bar{x}_1 + x_2x_1 + x_3x_2\bar{x}_1$$

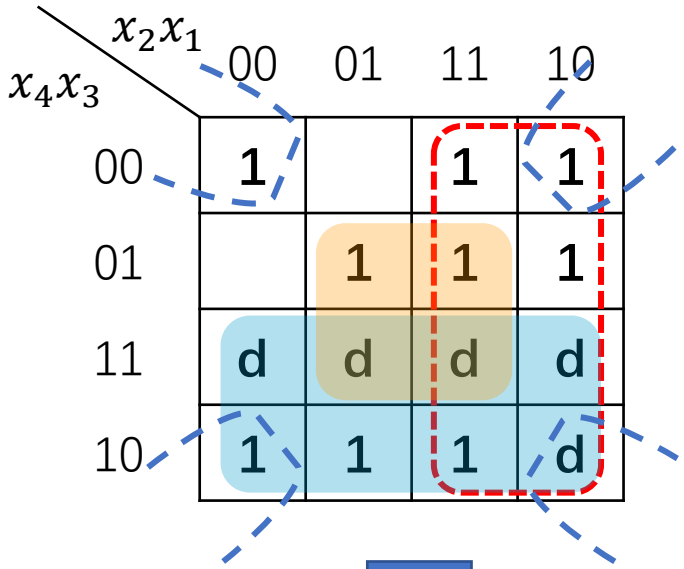
$$d = \bar{x}_3x_2 + x_3\bar{x}_2x_1 + \bar{x}_3\bar{x}_1 + x_3x_2\bar{x}_1$$

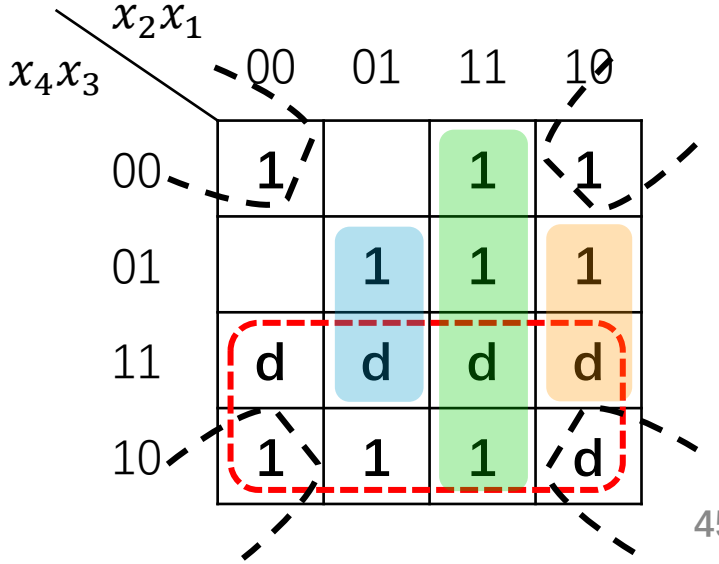
$$e = \bar{x}_3\bar{x}_1 + x_3x_2\bar{x}_1$$

$$f = x_3\bar{x}_2x_1 + \bar{x}_2\bar{x}_1 + x_3x_2\bar{x}_1 + x_4$$

$$g = x_3x_2\bar{x}_1 + x_3\bar{x}_2 + x_4 + \bar{x}_3x_2$$

多输出优化





兼顾结果：独立乘积项由15个减少到9个！

卡诺图小 结

- **卡诺图：**格雷码编码、相邻可合并、首尾相邻、单元可重复使用、横竖皆可
- 卡诺图中，相邻的最小项可以合并，并消去取值不同的因子
- **相邻**：最小项相差一个变量
- 被合并的相邻小方块数目越大，得到的乘积项中包含的字母变量越少
- **蕴含、质蕴含、必要质蕴含**
- 卡诺图化简结果不唯一
- 不要超过5个自变量
- 充分利用无关项
- 多输出函数要充分利用公共项